Title 0:00–0:15. Joint work with Ken Brown.

CSS codes and transversal gates

0:15–1:00 / 1 Outline Preview Steane and $[\![8,3,2]\!]$, conditions, geometry, completion, and bounds.

1:00–1:55 / 2 Binary vectors Qubits at cube points. Faces u_1, u_2. Overlap is an edge. XOR, not a direct sum.

1:55–2:55 / 3 Pauli strings $X(u_1)$ on a face. Overlap 2 commutes, overlap 1 anticommutes.

2:55–3:55 / 4 Stabilizer, CSS Syndrome, $[\![n,k,d]\!]$. $X(a)$ for $a \in C_2$, $Z(b)$ for $b \in C_1^{\perp}$. $G = [K;S]$ and $|x\rangle_L$.

3:55–4:35 / 5 Steane code Seven nonzero points. S is the faces, K is all ones. Terms u_1 and $\mathbf{1} \oplus u_1$.

4:35–5:40 / 6 Syndromes Z on u has syndrome u. Distinct columns detect pairs. A line is a weight-3 logical.

5:40–6:40 / 7 Clifford, non-Clifford Gates on basis states. Phase gate S, not the matrix. Add T or CCZ.

6:40–7:50 / 8 Transversal, distillation $H^{\otimes 7}$ turns the X face into a Z face: logical H. Eastin–Knill: no T. Distillation needs T or CCZ. State the question.

$[\![8,3,2]\!]$ **example** *7:50–8:00: outline, current theme highlighted.*

8:00–9:15 / 9 $[\![8,3,2]\!]$ code Origin back, roles swapped. Pairs for $x = 100, 111$. Edge: d_Z=2, d_X=4.

9:15–10:15 / 10 Signed weights Integer signed weight, mod 8. Every codeword in a logical coset needs the same phase.

10:15–12:15 / 11 Pattern from overlaps Ask where $T^{\dagger}$ goes. Reveal $\gamma = u_1 \oplus u_2 \oplus u_3$. Face 0, edge 0, vertex -1.

12:15–13:00 / 12 Row congruences Coefficients 1, -2, 4 give mod 8, 4, 2. Only the logical triple is odd.

Gate conditions *13:00–13:10: outline, current theme highlighted.*

13:10–14:10 / 13 Triorthogonal codes Timeline. NH22 range is $n + k$, not our n. Even overlaps, but CCZ needs one odd triple.

14:10–15:40 / 14 Nine CH conditions Logical, mixed, stabilizer families. K3 odd. Parities allow a Clifford correction.

Geometric construction *15:40–15:50: outline, current theme highlighted.*

15:50–17:10 / 15 Q48 result 48 physical, 3 logical. d_X=16, d_Z=3. 26 T and 22 inverse, no correction.

17:10–18:10 / 16 Fix the stabilizers Projective columns. 8-divisible span. Find K. Solve signs. Divisibility is a choice.

18:10–19:10 / 17 Projective syndromes Nonzero columns detect one Z. Distinct columns detect two. This proves only d_Z.

19:10–20:10 / 18 48-point support 63 minus 15 gives 48. Remove $w = 00$. Three blocks of 16. Six coordinate rows.

20:10–21:40 / 19 Weights 24 and 32 Nonzero u coefficient: 3 times 8. Pure w coefficient: 2 times 16. Rank six.

21:40–22:40 / 20 Geometric family Recover 48 at $p = 4, q = 2$. Other counts are lower bounds. Star is a fixed-support census.

Column picture and completion *22:40–22:50: outline, current theme highlighted.*

22:50–24:10 / 21 Column indicators Occupied columns as a Boolean function. $[\![8,3,2]\!]$: y. Q48: $w_1 + w_2 + w_1 w_2$.

24:10–25:40 / 22 Moments and monomials Overlap parity is a monomial moment. One exceptional cubic. State the coset briefly.

25:40–27:10 / 23 Logical labels Three label bits per point. Product forces $x = \kappa$. S-only constraints already hold.

27:10–29:10 / 24 Logical completion One-row space, compatible pairs, linear third row. Three-site witness makes d_Z=3.

29:10–30:40 / 25 $T/T^{\dagger}$ signs Binary gamma in mod-4 equations. Show the polynomial, count $6 + 6 + 10$ inverse gates.

Bounds and questions *30:40–30:50: outline, current theme highlighted.*

30:50–32:40 / 26 Reed–Muller supports Positive-degree tests vanish at 0. Correct weight parity, then use RM duality.

32:40–34:10 / 27 Exclusion through 38 Rank at most 8, support reductions, complete enumeration, no logical completion.

34:10–36:10 / 28 Length comparison Our 38 floor. WJS 39–46. 47 quasi-transversal. Sign failure is code-specific.

36:10–37:40 / 29 Takeaways and questions Geometry, labels, signs. Ask about higher distance and sign feasibility. Leave QR up.

37:40: finish and invite questions.

Q&A / B1–B4 Backups Reed–Muller lengths, rank reduction, finite cases, dictionary with [CLM+26].